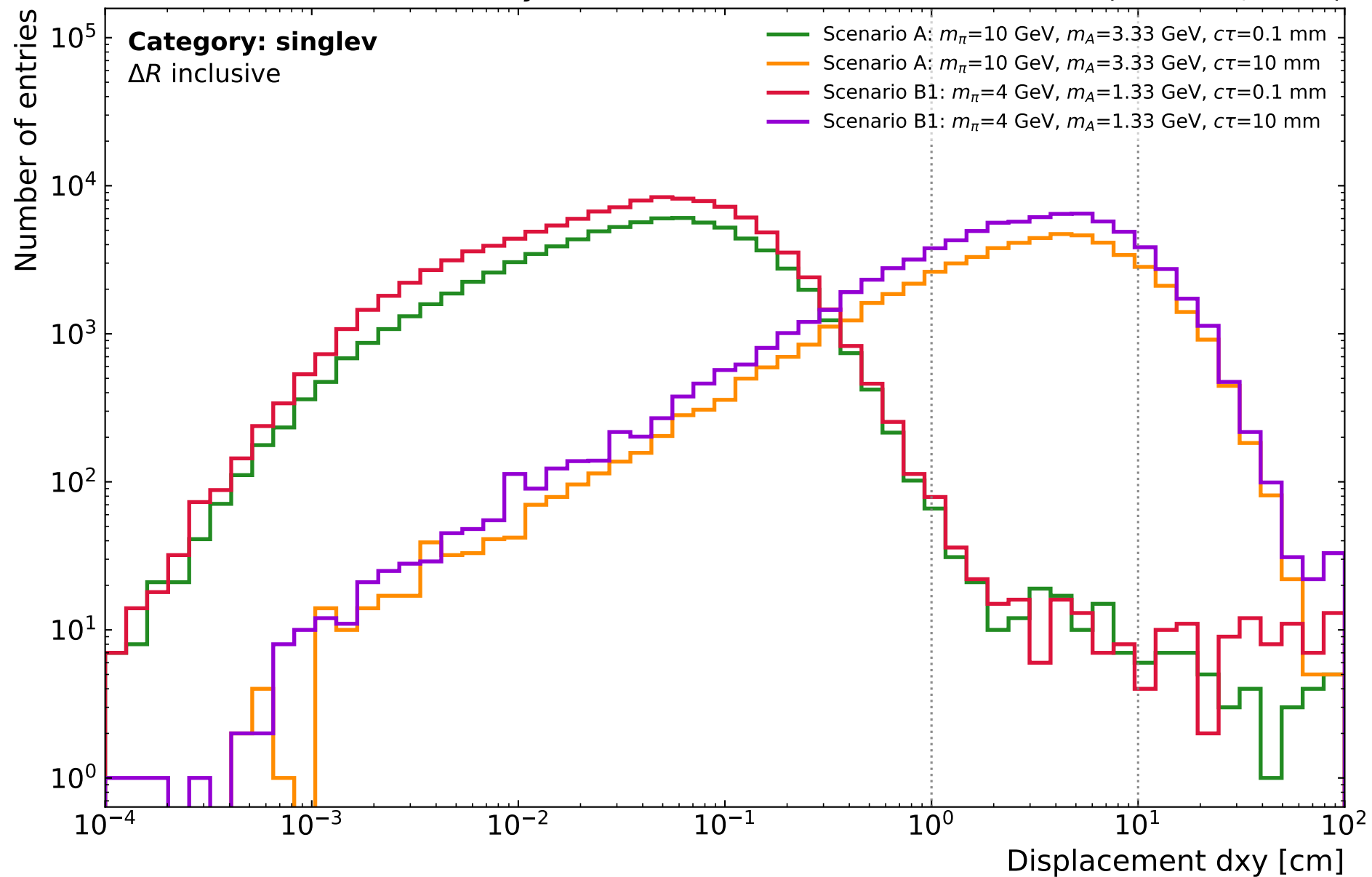


singlev: displacement of the binning vertex (muonSV\_dxy at min\_chi2\_index)  
 $\Delta R$  inclusive; dotted: dxy category edges (1, 10 cm)

**CMS** *Simulation Preliminary*

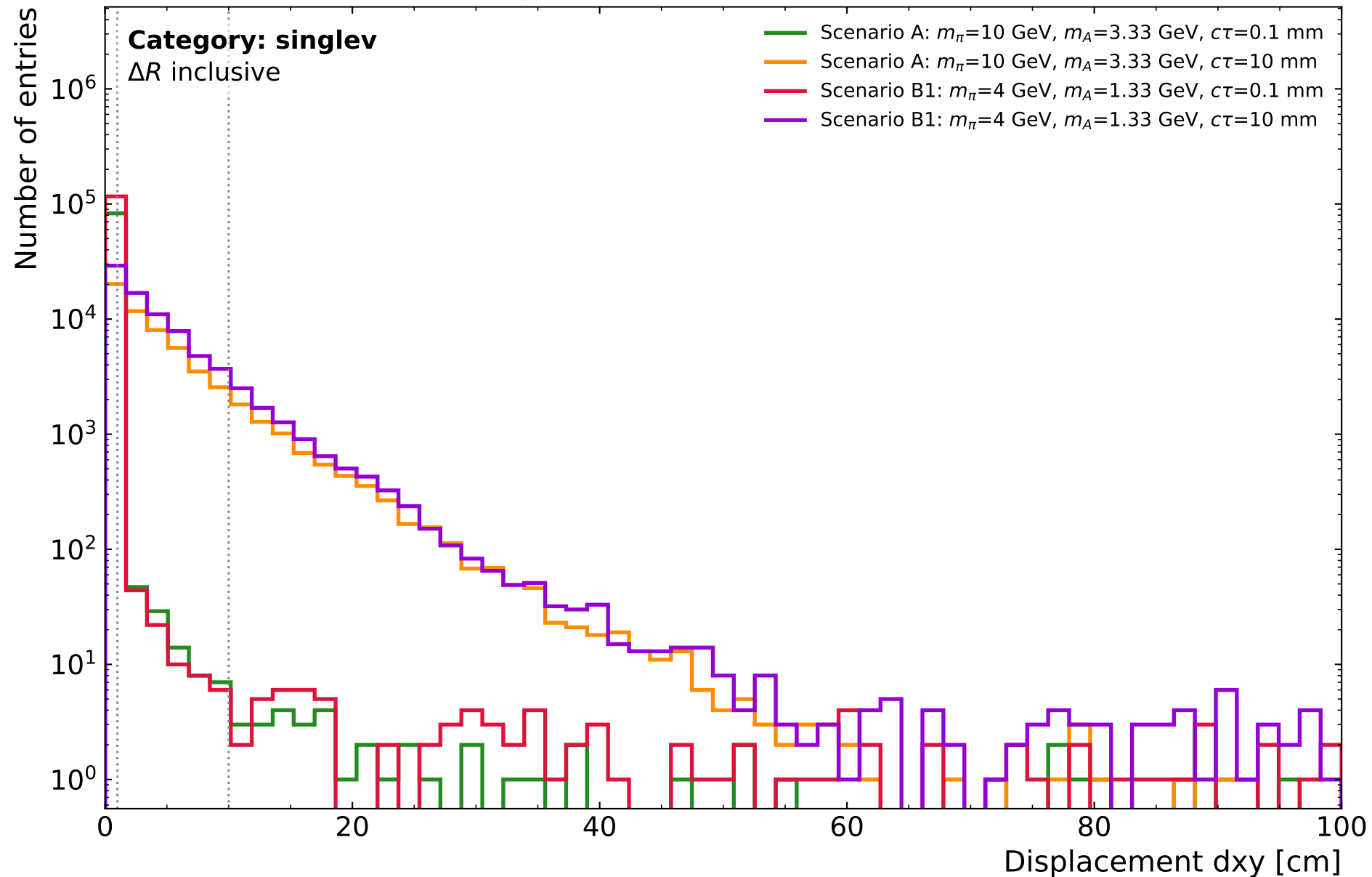
(13.6 TeV, 2024)



singlev: displacement of the binning vertex (muonSV\_dxy at min\_chi2\_index)  
 $\Delta R$  inclusive; dotted: dxy category edges (1, 10 cm)

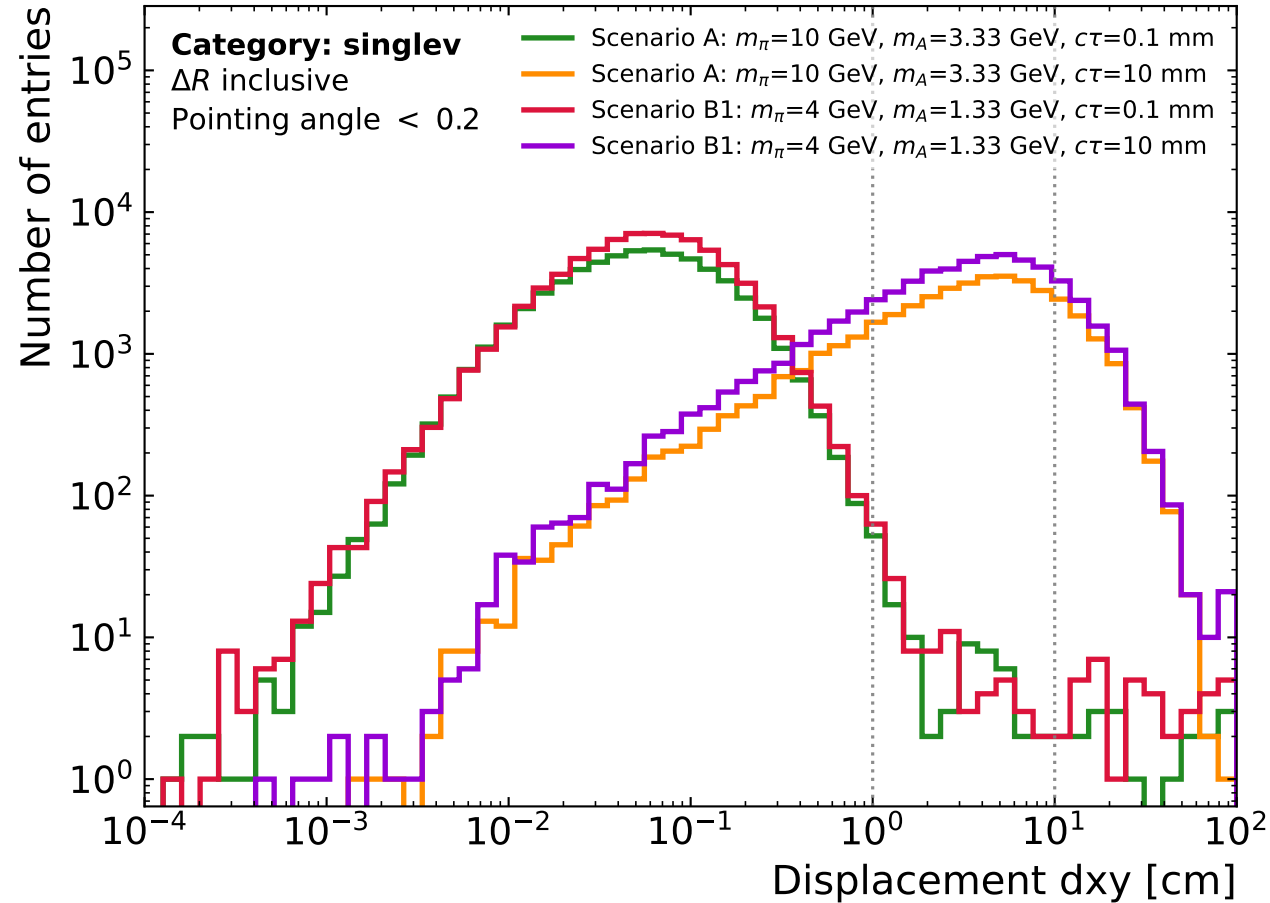
**CMS** *Simulation Preliminary*

(13.6 TeV, 2024)



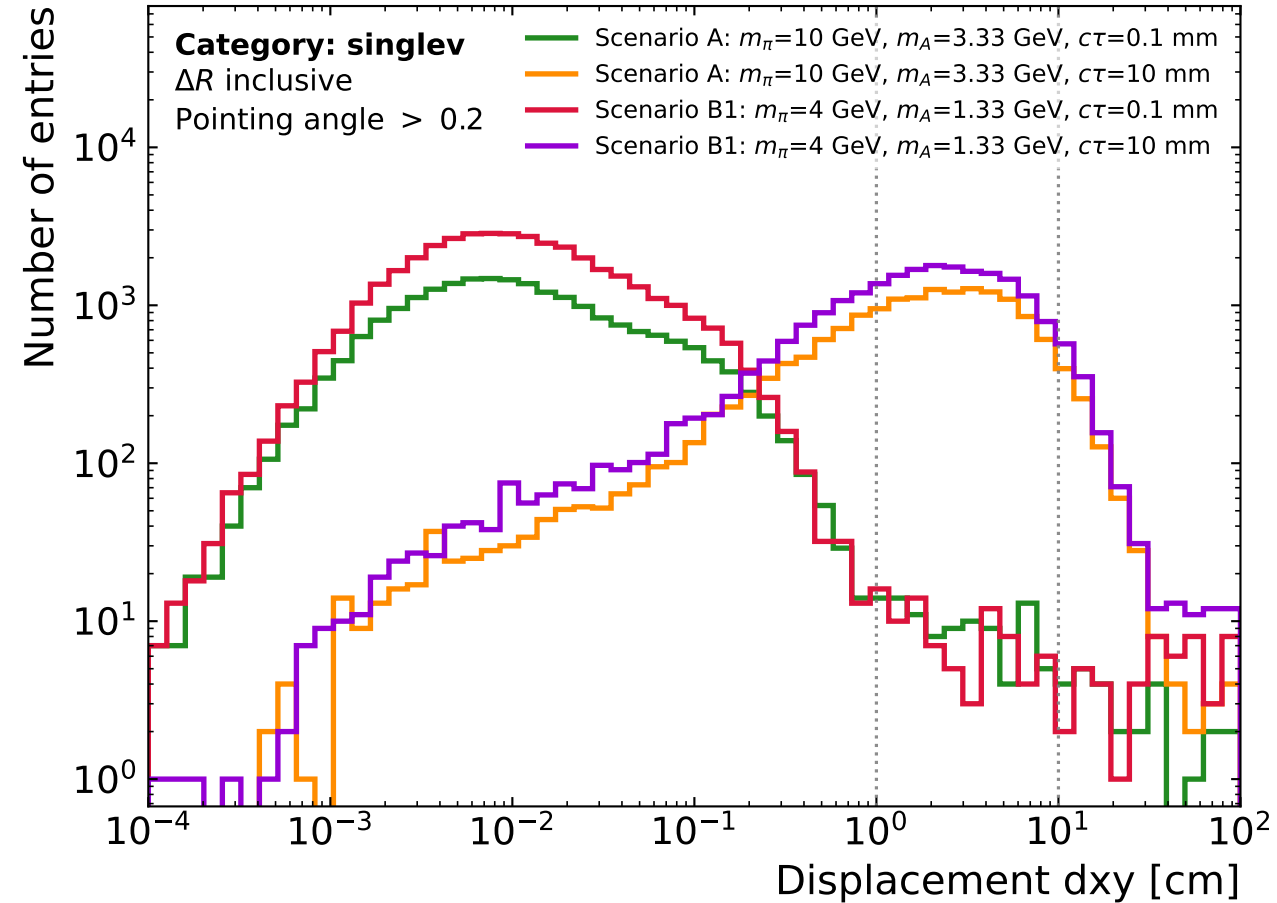
singlev: displacement of the binning vertex (muonSV\_dxy at min\_chi2\_index), split by pointing angle  
 $\Delta R$  inclusive; dotted: dxy category edges (1, 10 cm)

**CMS Simulation Preliminary** (13.6 TeV, 2024)



|                  | dxy < 1 | 1--10  | dxy > 10 | Total  |
|------------------|---------|--------|----------|--------|
| scA $c\tau=0.1$  | 60,450  | 85     | 18       | 60,553 |
| scA $c\tau=10$   | 8,169   | 27,401 | 6,641    | 42,211 |
| scB1 $c\tau=0.1$ | 75,257  | 103    | 41       | 75,401 |
| scB1 $c\tau=10$  | 11,847  | 39,135 | 8,457    | 59,439 |

**CMS Simulation Preliminary** (13.6 TeV, 2024)

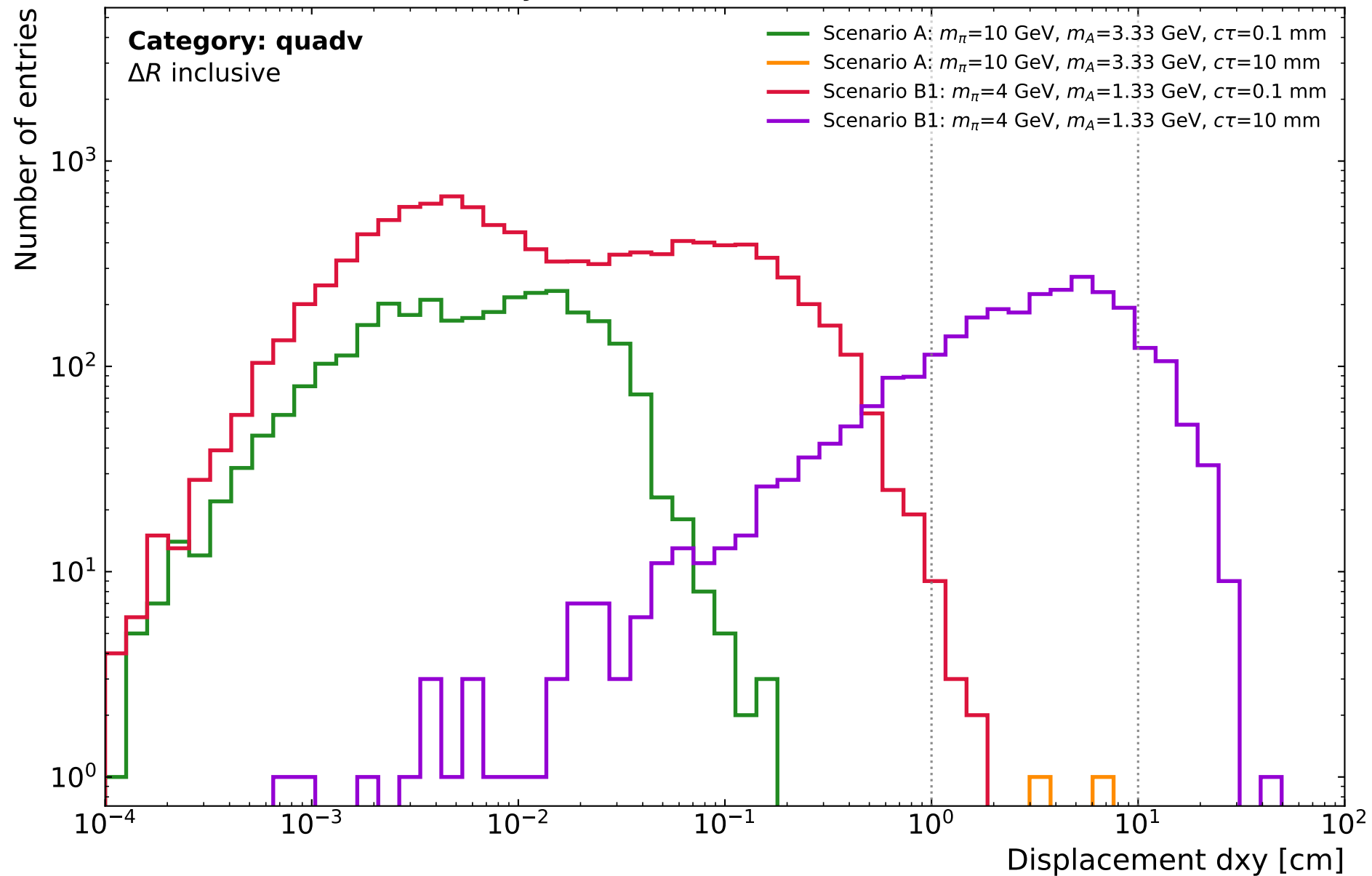


|                  | dxy < 1 | 1--10  | dxy > 10 | Total  |
|------------------|---------|--------|----------|--------|
| scA $c\tau=0.1$  | 22,371  | 92     | 27       | 22,490 |
| scA $c\tau=10$   | 5,347   | 10,450 | 812      | 16,609 |
| scB1 $c\tau=0.1$ | 41,004  | 80     | 50       | 41,134 |
| scB1 $c\tau=10$  | 7,600   | 14,441 | 1,133    | 23,174 |

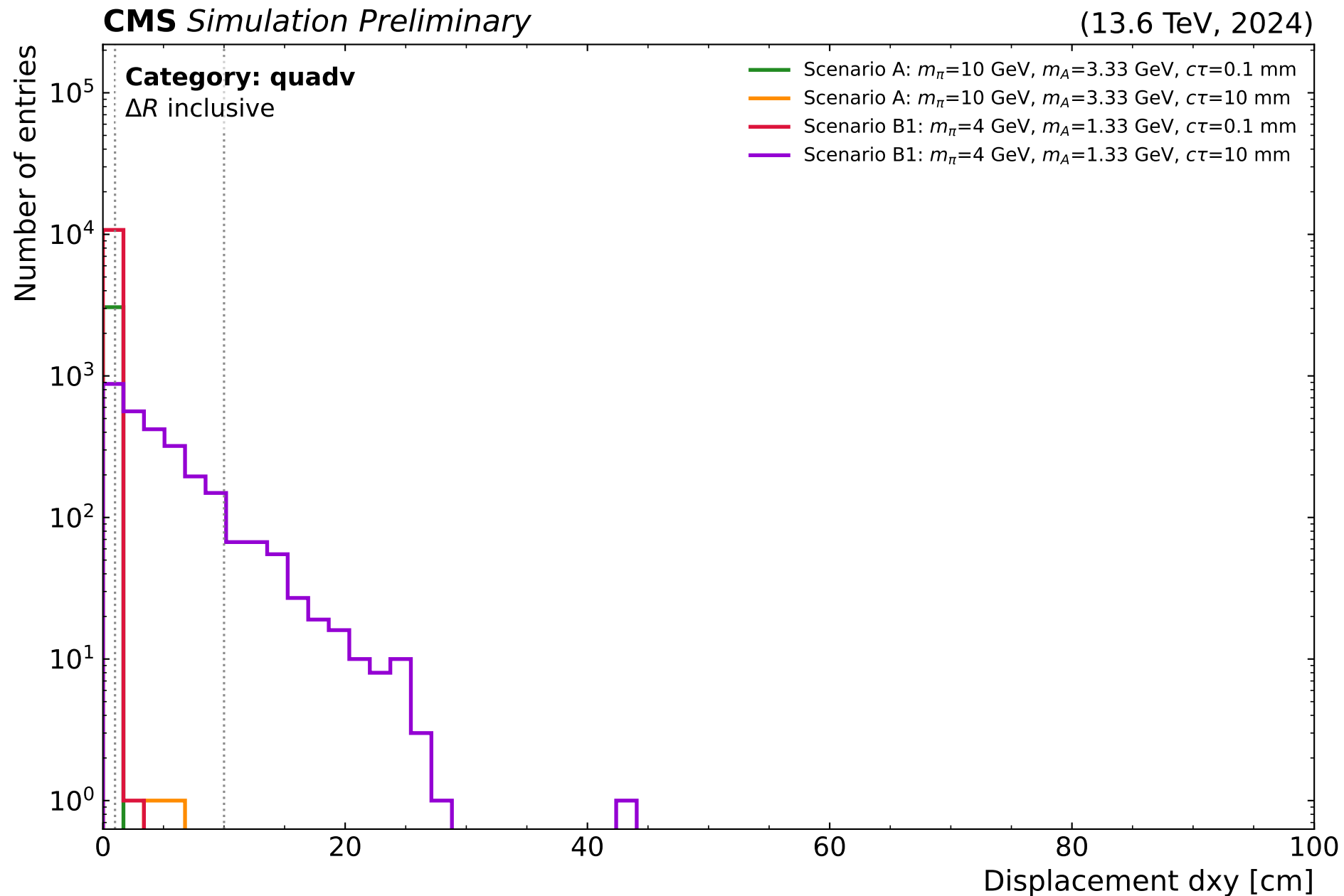
quadv: displacement of the binning vertex (fourmuonSV\_dxy)  
 $\Delta R$  inclusive; dotted: dxy category edges (1, 10 cm)

**CMS** *Simulation Preliminary*

(13.6 TeV, 2024)

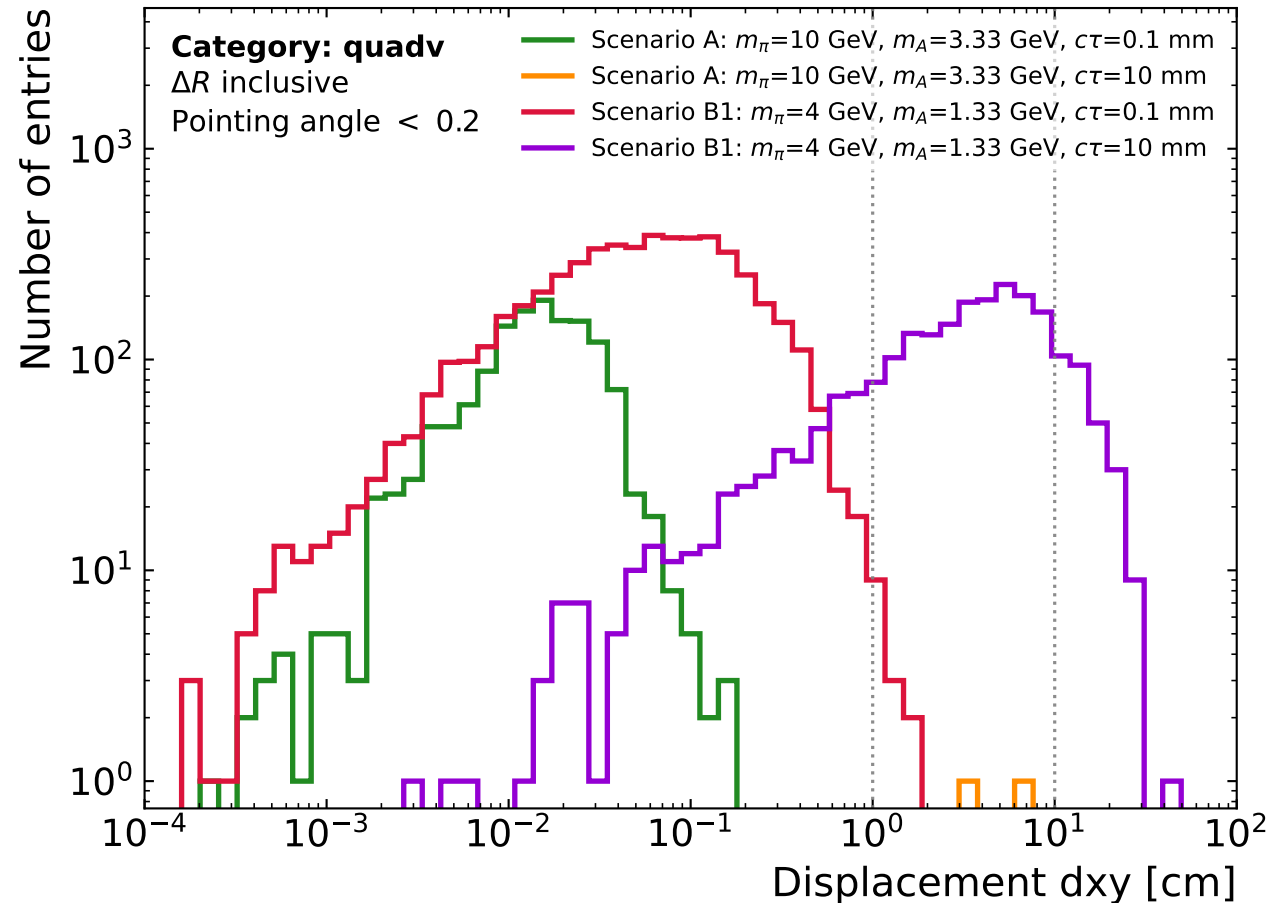


quadv: displacement of the binning vertex (fourmuonSV\_dxy)  
 $\Delta R$  inclusive; dotted: dxy category edges (1, 10 cm)



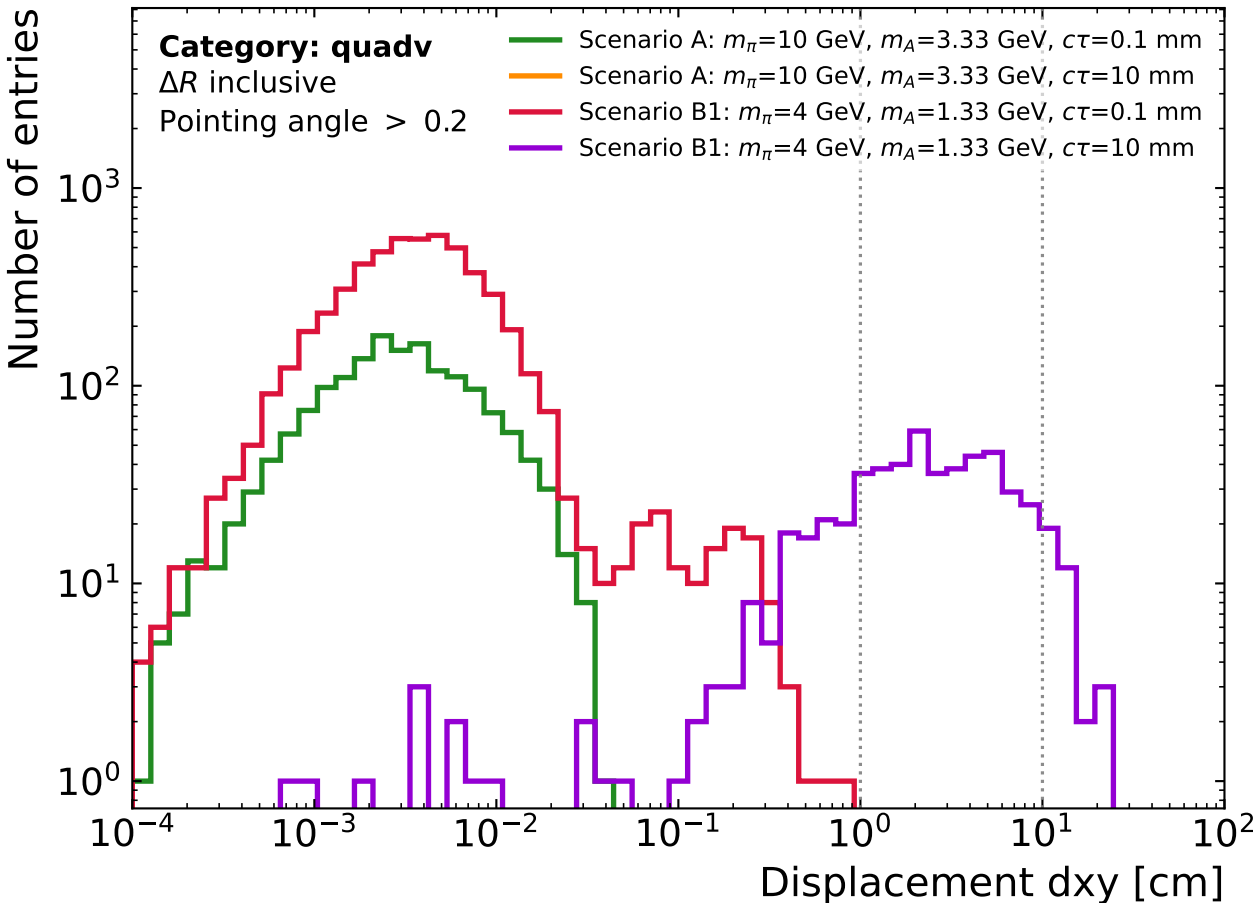
quadv: displacement of the binning vertex (fourmuonSV\_dxy), split by pointing angle  
 $\Delta R$  inclusive; dotted: dxy category edges (1, 10 cm)

**CMS Simulation Preliminary** (13.6 TeV, 2024)



|                  | dxy < 1 | 1--10 | dxy > 10 | Total |
|------------------|---------|-------|----------|-------|
| scA $c\tau=0.1$  | 1,403   | 0     | 0        | 1,403 |
| scA $c\tau=10$   | 0       | 2     | 0        | 2     |
| scB1 $c\tau=0.1$ | 5,338   | 11    | 0        | 5,349 |
| scB1 $c\tau=10$  | 449     | 1,553 | 267      | 2,269 |

**CMS Simulation Preliminary** (13.6 TeV, 2024)

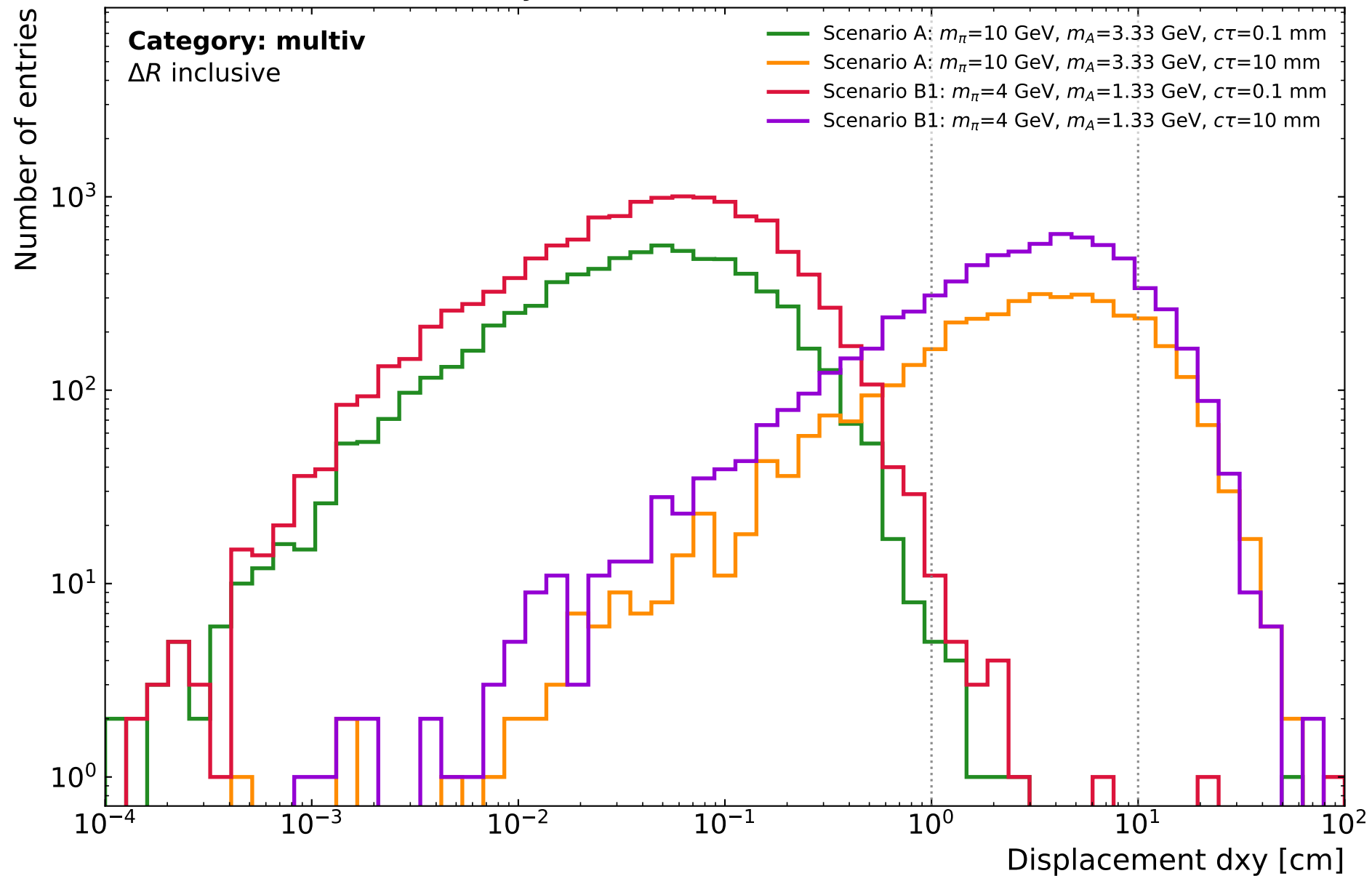


|                  | dxy < 1 | 1--10 | dxy > 10 | Total |
|------------------|---------|-------|----------|-------|
| scA $c\tau=0.1$  | 1,656   | 0     | 0        | 1,656 |
| scA $c\tau=10$   | 0       | 0     | 0        | 0     |
| scB1 $c\tau=0.1$ | 5,402   | 0     | 0        | 5,402 |
| scB1 $c\tau=10$  | 123     | 386   | 30       | 539   |

multiv: displacement of the binning vertex (muonSV\_dxy at min\_chi2\_index)  
 $\Delta R$  inclusive; dotted: dxy category edges (1, 10 cm)

**CMS Simulation Preliminary**

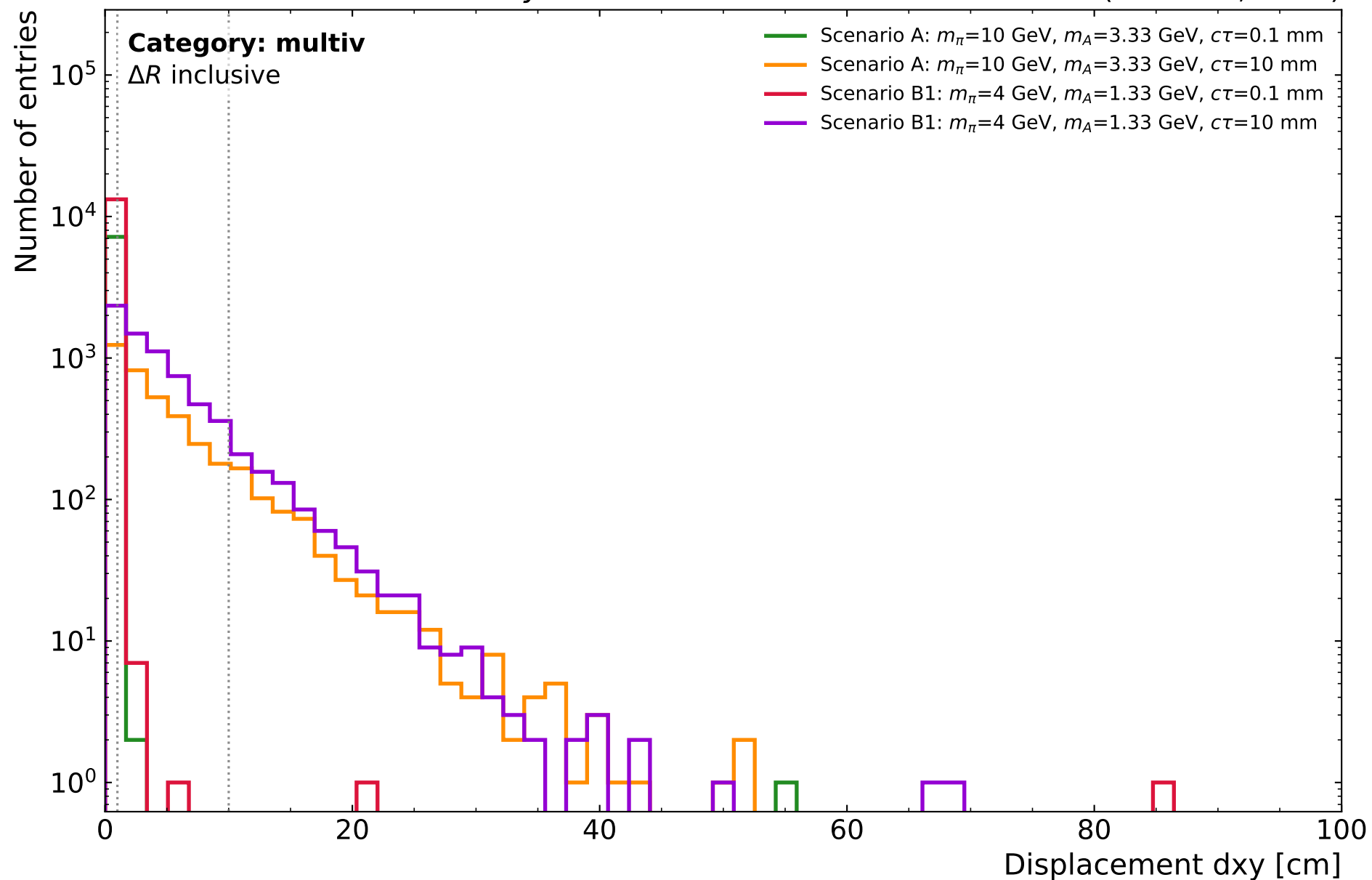
(13.6 TeV, 2024)



multiv: displacement of the binning vertex (muonSV\_dxy at min\_chi2\_index)  
 $\Delta R$  inclusive; dotted: dxy category edges (1, 10 cm)

**CMS** *Simulation Preliminary*

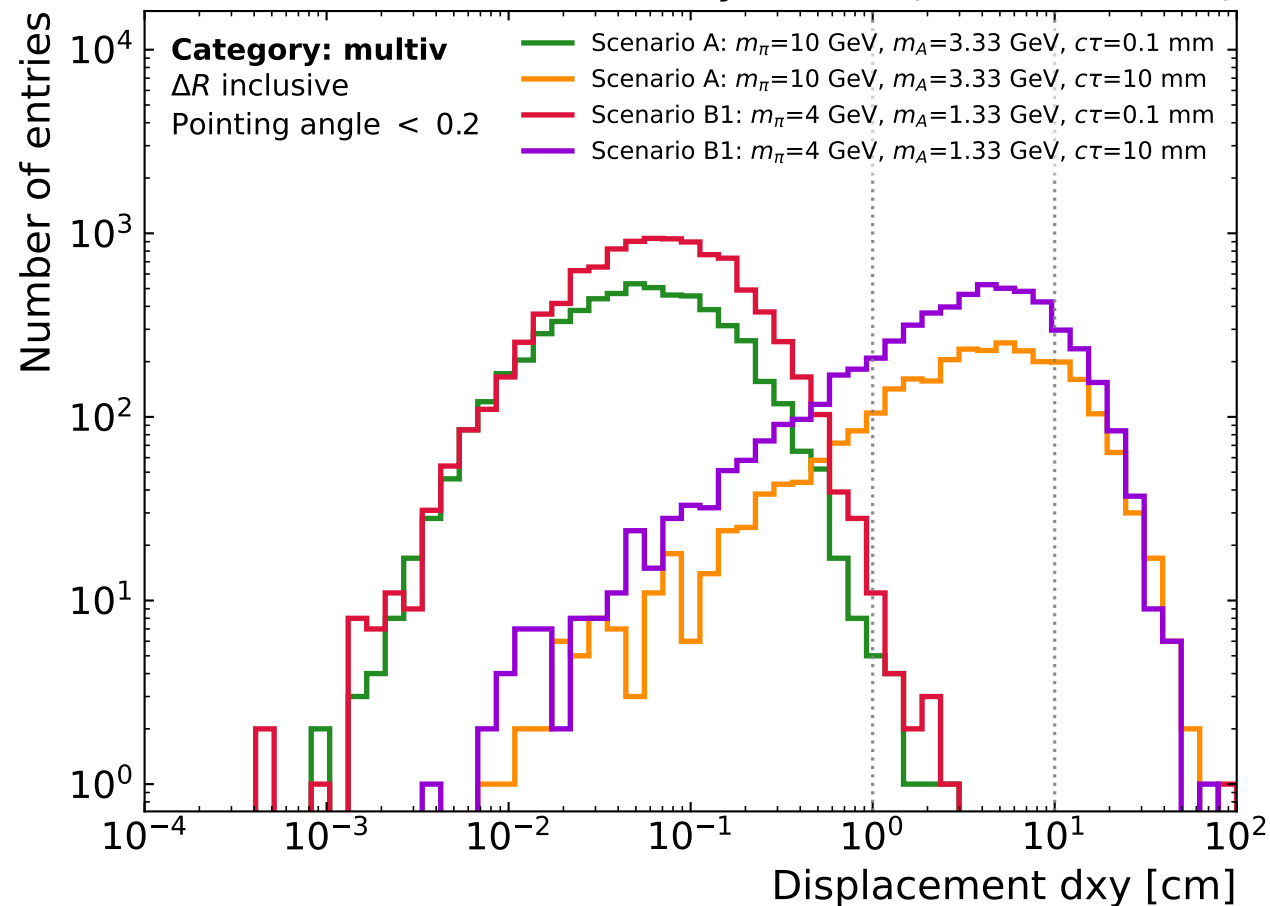
(13.6 TeV, 2024)





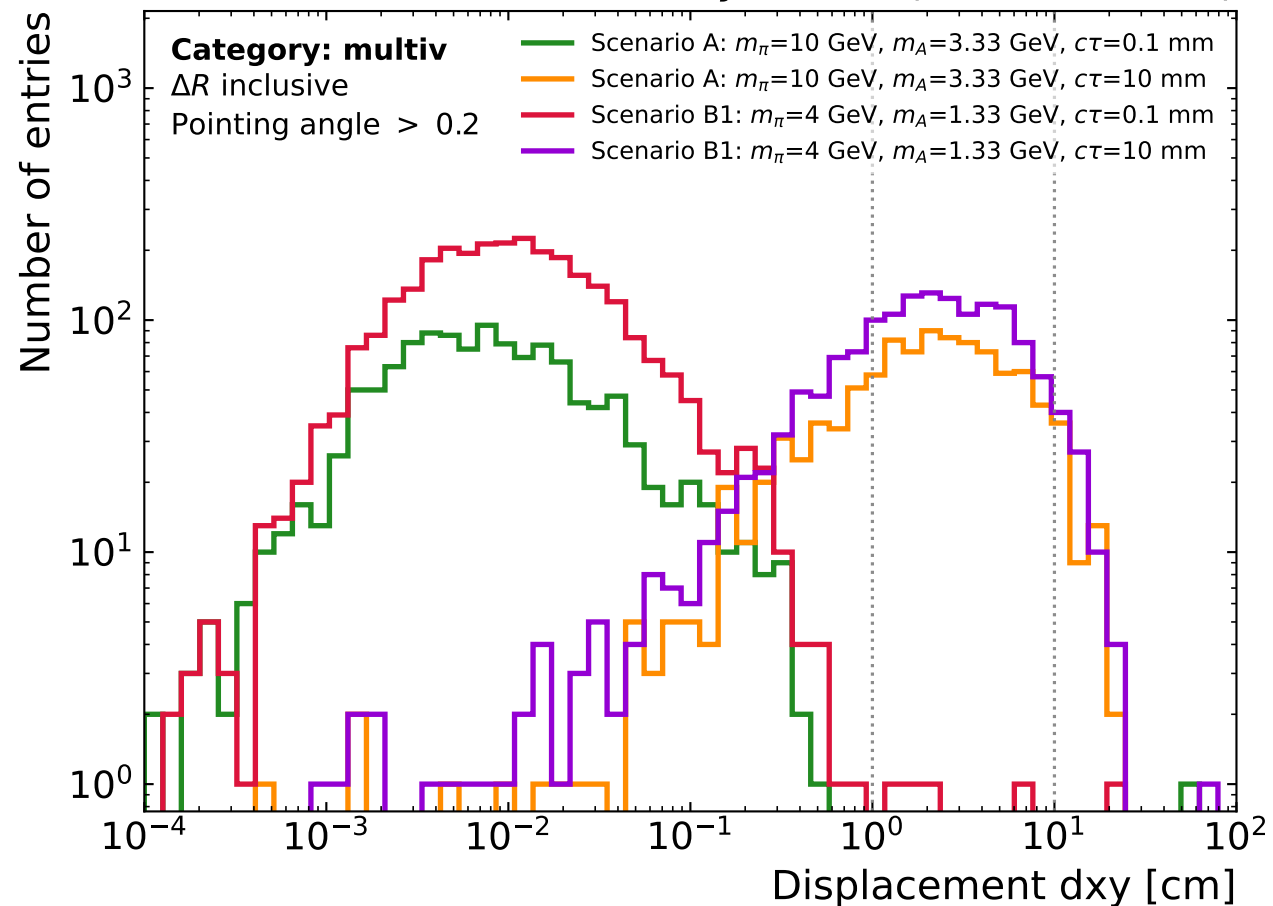
multiv: displacement of the binning vertex (muonSV\_dxy at min\_chi2\_index), split by pointing angle  
 $\Delta R$  inclusive;    dotted: dxy category edges (1, 10 cm)

**CMS Simulation Preliminary** (13.6 TeV, 2024)



|                  | dxy < 1 | 1--10 | dxy > 10 | Total  |
|------------------|---------|-------|----------|--------|
| scA $c\tau=0.1$  | 5,926   | 9     | 0        | 5,935  |
| scA $c\tau=10$   | 501     | 1,922 | 547      | 2,970  |
| scB1 $c\tau=0.1$ | 10,248  | 18    | 1        | 10,267 |
| scB1 $c\tau=10$  | 1,107   | 3,923 | 761      | 5,791  |

**CMS Simulation Preliminary** (13.6 TeV, 2024)



|                  | dxy < 1 | 1--10 | dxy > 10 | Total |
|------------------|---------|-------|----------|-------|
| scA $c\tau=0.1$  | 1,248   | 0     | 1        | 1,249 |
| scA $c\tau=10$   | 280     | 685   | 55       | 1,020 |
| scB1 $c\tau=0.1$ | 2,962   | 4     | 1        | 2,967 |
| scB1 $c\tau=10$  | 425     | 1,039 | 72       | 1,536 |